

EITT + CoDa: Complete Study & Position Sheet

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1. Glossary of Terms — GROUPED BY ORIGIN

CoDa STANDARD TERMS (Aitchison, Egozcue, Pawlowsky-Glahn et al. — the mathematics)

These are established terms from the Compositional Data Analysis literature. HUF did not invent them.

Term	Symbol / Formula	What It Means
Aitchison distance	$d_A(x,y) = \ \text{clr}(x) - \text{clr}(y)\ _2$	The natural ruler on the simplex. Measures how different two compositions are in log-ratio space.
Aitchison geometry	$(S^D, (+), (.), <, >_A)$	The Hilbert space structure of the simplex: perturbation, powering, inner product.
Aitchison norm	$\ x\ _A = d_A(x, n)$	Distance from the uniform composition. Measures concentration.
ALR (Additive Log-Ratio)	$\text{alr}(x)_i = \ln(x_i/x_D)$	All parts vs one reference. Simple but NOT isometric. Depends on which part chosen.
Balance (ILR coordinate)	$y_k = \sqrt{rs/(r+s)} \ln(g(G^+)/g(G^-))$	A single ILR coordinate. Normalised log-ratio of geometric means of two groups.
CLR (Centered Log-Ratio)	$\text{clr}(x)_i = \ln(x_i/g(x))$	Each part vs geometric mean of all. Symmetric but singular (D values, D-1 dims).
Closure	$C(z) = (z_i / \sum z_j)$	Force any vector to sum to 1 (or kappa). The gate onto the simplex.
CoDa	Compositional Data Analysis	The field. Parts of a whole. Founded by Aitchison (1986).
Composition	$x = (x_1, \dots, x_D)$ in S^D	D positive numbers summing to 1. One point on the simplex.
Frechet mean	$\bar{x} = \text{argmin} \sum d_A(x, x(t))^2$	The "average" minimising total Aitchison distance. = geometric mean on the simplex.
Geometric mean (of comps)	$\bar{x}_G = C(\text{Prod } x_i(t)^{1/N})$	Per-part geometric mean, re-close. Frechet mean on the simplex.
ILR (Isometric Log-Ratio)	$y = \Psi^T \cdot \text{clr}(x)$ in R^{D-1}	Gold standard transform. Preserves distances, full-rank, unconstrained.
Isometry	$d_A(x,y) = \ \text{ilr}(x) - \text{ilr}(y)\ _2$	ILR preserves Aitchison distances as Euclidean distances. No distortion.
Neutral element	$n = (1/D, \dots, 1/D)$	Uniform composition. Max entropy state. "No opinion" on where energy belongs.
Pairwise log-ratio	$\lambda_{ij} = \ln(x_i/x_j)$	Exchange rate between parts i and j. The fundamental atom of CoDa.
Perturbation	$x (+) p = C(x_1 p_1, \dots, x_D p_D)$	Simplex "addition". Multiply parts by factors, re-close.
Powering	$a (.) x = C(x_1^a, \dots, x_D^a)$	Simplex "scalar multiplication". Amplifies or compresses all contrasts.
SBP (Sequential Binary Partition)	Tree of binary splits on D parts	The wiring diagram for ILR balances. Choice does not affect total variance.
Shannon entropy	$H(x) = -\sum x_i \ln x_i$	Diversity/evenness. $H=0$ at monoculture, $H=\ln(D)$ at uniform. Standard info theory.
Simplex	$S^D = \{x: x_i > 0, \sum x_i = 1\}$	The sample space for compositions. A (D-1)-dimensional constrained surface.
Subcomposition	$x_S = C(x_i : i \text{ in } S)$	Pick subset of parts, re-close. Like looking at only fossil fuels.
Subcomp. coherence	Results on sub = results on full	Valid CoDa operations give same answer on any subcomposition. Aitchison has this.
Total Aitchison variance	$\text{totvar} = \sum \text{Var}(y_k)$	Total spread. = sum of ILR balance variances. SBP-invariant.

HUF TERMS (Higgins, Rogue Wave Audio — the monitoring framework)

These are terms introduced by HUF. They use CoDa mathematics but the concepts are new.

Term	Symbol / Formula	What It Means
Adaptive decimation	D_{adapt}	Variable-width block averaging. $d(\text{CoDa})/dt$ sets the block width. Compress more where slow, less where fast.

Autocorrelation boundary	3 tests: lag-1, block var ratio, entropy drift sign	The stationarity check. If these fail, data changes too fast for EITT at that ratio.
Balance velocity	$v_k(t) = dy_k/dt $	How fast a balance is changing. The derivative of one ILR coordinate.
Block decimation	$D_M[\{x(t)\}] = \{x\text{-bar}_G^{(b)}\}$	Chop time series into blocks of M, replace each with geometric mean. Core EITT operation.
d(CoDa)/dt	$V(t) = \max_k dy_k/dt $	Max balance velocity across all taps. The tap controller. Determines safe averaging window.
Deceptive drift	K_{eff} drops while TV stays calm	Concentration building behind apparent stability. The silent failure MC-4 detects.
EITT	Entropy Invariance under Temporal Transformation	Core finding: Shannon entropy barely changes under geometric-mean decimation.
EITT Inversion	K fails -> increase K -> passes	When EITT fails, increasing K restores it. Failure diagnoses hidden structure.
EITT residual	$\text{delta}_M = [\text{mean } H(\text{dec}) - \text{mean } H(\text{orig})] / \text{mean } H(\text{orig}) \times 100\%$	% change in mean entropy after decimation. Small = invariance holds. Typically <1%.
HUF-GOV	Open-loop governance	Measure, report, file. No intervention. The scientific instrument. Stateless.
HUF-CLS	Closed-loop system	Measure, interpret, adjust, re-measure. Feedback. Licensed through RWA.
Isotropic Principle	Ground state = $(1/D, \dots, 1/D)$	The geometry exists before the signal. Every departure from isotropy is a signal.
K_{eff} (effective carriers)	$K_{\text{eff}} = \exp(H(x))$	How many carriers are "effectively present". = perplexity in NLP, Hill number $q=1$.
MC-4	Composition Monitoring	The fourth monitoring category (after Magnitude, Identity, Trend). No ISO standard covers it.
Multi-tap test	8 SBP trees x 3 datasets x 3 ratios	Test EITT across ALL valid SBP trees. 94.8% pass. Geometry property, not wiring artefact.
TV distance diagnostic) (as	$d_{TV} = 0.5 \text{ Sum } x_i - y_i $	L1 on raw shares. NOT simplex-native. Used as comparison diagnostic alongside d_A .

2. Key CoDa Theories and Who Produced Them

The mathematical foundations HUF stands on. Every one of these predates HUF.

Theory / Result	Who	When	What It Established
Log-ratio approach to compositional data	John Aitchison	1982, 1986	Founded CoDa. Showed Euclidean statistics on the simplex are invalid. Introduced log-ratio transforms and the Aitchison metric.
ILR transform and Hilbert space structure	Egozcue, Pawlowsky-Glahn, Mateu-Figueras, Barcelo-Vidal	2003	Proved the simplex is a Hilbert space. Built the isometric log-ratio (ILR) transform. The gold standard.
Aitchison inner product and geometry	Pawlowsky-Glahn, Egozcue	2001	Formalised the inner product, norm, and distance on the simplex. Made CoDa a proper Euclidean vector space via ILR.
Textbook: "Modeling and Analysis of CoDa"	Pawlowsky-Glahn, Egozcue, Tolosana-Delgado	2015	The definitive reference. Unifies 30 years of CoDa theory. Covers SBP, balances, biplots, regression, multivariate analysis.
Hill numbers and diversity unification	Hill (1973); Chao, Jost (2012); Chao et al. (2014, iNEXT)	1973-2014	$K_{eff} = \exp(H)$ IS Hill number of order 1. Unified Simpson, Shannon-Wiener, species richness into one family.
Subcompositional coherence principle	Aitchison (1986); Egozcue, Pawlowsky-Glahn (2003)	1986-2003	Results must be invariant to which subset of parts you examine. Aitchison distance has this. Euclidean does not.
Perturbation algebra on the simplex	Aitchison (1986); Barcelo-Vidal, Mateu-Figueras (2001)	1986-2001	The simplex forms an Abelian group under perturbation. Perturbation difference is the compositional analogue of subtraction.
SBP and balance decomposition	Egozcue, Pawlowsky-Glahn (2005)	2005	Any valid binary partition of D parts produces D-1 orthonormal balances. Total variance is SBP-invariant. The wiring theorem.
Fisher-Rao metric on the simplex	Amari (1985); Erb, Ay (2020)	1985-2020	The simplex as a statistical manifold. Chentsov uniqueness theorem. Connects CoDa to information geometry.
Shannon entropy and information theory	Shannon (1948)	1948	$H(x) = -\sum x_i \ln x_i$. The most studied formula in information science. EITT says it is near-invariant under CoDa temporal decimation.

3. Formula Reference — Grouped

CoDa SIMPLEX OPERATIONS (standard)

Operation	Formula	Analogy
Closure	$C(z) = (z_i / \sum z_j)$	Normalise to 100%
Perturbation (+)	$x (+) p = C(x_1 p_1, \dots, x_D p_D)$	Gain adjustment
Pert. difference (-)	$x (-) y = C(x_1/y_1, \dots, x_D/y_D)$	Compositional first-difference
Powering (.)	$a (.) x = C(x_1^a, \dots, x_D^a)$	Contrast amplifier
Geometric mean	$g(x) = (\text{Prod } x_i)^{1/D}$	The CoDa "center"
Neutral element	$n = (1/D, \dots, 1/D)$	Flat signal / max entropy

CoDa LOG-RATIO TRANSFORMS (standard)

Transform	Formula	Dim	Use
CLR	$\text{clr}(x)_i = \ln(x_i/g(x))$	R^D (sing.)	Visualisation, biplots
ILR	$y = \text{Psi}^T \cdot \text{clr}(x)$	R^{D-1} (full)	Formal statistics, gold standard
ILR balance	$y_k = \sqrt{\text{rs}/(\text{r}+\text{s})} \ln(g(G^+)/g(G^-))$	Scalar	Each tap on the transformer
ALR	$\text{alr}(x)_i = \ln(x_i/x_D)$	R^{D-1} (not iso)	Quick exploration only

HUF / EITT-SPECIFIC FORMULAS (new work, Higgins / Rogue Wave Audio)

Formula	Definition	Status
Block decimation	D_M : replace each block of M with $x\text{-bar}_G = C(\text{Prod } x_i(t)^{1/M})$	Definition
CLR identity	$\text{clr}(x\text{-bar}_G) = (1/M) \sum \text{clr}(x(t))$ over block	Mathematical
ILR identity	$\text{ilr}_k(x\text{-bar}_G) = (1/M) \sum \text{ilr}_k(x(t))$ over block	Proved (10^{-16})

EITT residual	$\text{delta}_M = [\text{mean } H(\text{dec}) - \text{mean } H(\text{orig})] / \text{mean } H(\text{orig}) \times 100\%$	Empirical <2%
Hessian footprint	$\text{delta } H \sim (1/2) \text{tr}[\text{Hess } H \cdot \text{Cov}(\bar{x}_{G_i})]$, $\text{Hess}_{ii} = -1/x_i$	Approx. bound
Balance mean shift	$\Delta y\text{-bar}_k = \text{mean}(\text{ilr}_k \text{ after } D_M) - \text{mean}(\text{ilr}_k \text{ before})$	<0.1 at M=2
Balance velocity	$v_k(t) = y_k(t+1) - y_k(t-1) / 2$	Operationalised
Max rate (tap controller)	$V(t) = \max_k v_k(t)$	Operationalised
Adaptive block width	$M(t) = M_{\max}$ if $V(t) \leq \theta_L$; 2 if $V(t) > \theta_H$; interpolate	Operationalised

CoDa / INFO THEORY DISTANCES (who owns what)

Metric	Formula	Origin
Aitchison distance	$d_A = \ \text{clr}(x) - \text{clr}(y)\ _2$	CoDa (Aitchison)
Aitchison inner product	$\langle x, y \rangle_A = (1/D) \sum \ln(x_i/x_j) \ln(y_i/y_j)$	CoDa (Pawlowsky-Glahn)
Shannon entropy	$H(x) = -\sum x_i \ln x_i$	Info theory (Shannon 1948)
K_{eff}	$K_{\text{eff}} = \exp(H(x))$	Ecology (Hill 1973) / HUF naming
TV distance	$d_{TV} = 0.5 \sum x_i - y_i $	Standard (NOT simplex-native)
Total Aitchison variance	$\text{totvar} = \sum \text{Var}(y_k)$	CoDa (Egozcue 2003)

4. Results: Three Layers

LAYER 1 — MATHEMATICAL (proved to machine precision)

Result	Statement	Why
Balance preservation	mean $\text{ilr}_k(\bar{x}_G) = (1/M) \sum \text{ilr}_k(x(t))$ exactly	ILR is linear isometry; geom mean = arith mean in CLR
SBP invariance	Holds for ALL valid SBP trees simultaneously	Total ILR var = total Aitchison var, independent of SBP
Log-ratio means	Mean of $\ln(x_i/x_j)$ preserved under decimation	$\lambda_{ij} = \text{clr}_i - \text{clr}_j$, linear in CLR
Variance reduction	$\text{Var}(y_k)$ shrinks, ratio is SBP-invariant (spread <0.006)	Block averaging reduces variance; isometry preserves ratios

LAYER 2 — EMPIRICAL (observed, not proved)

Result	Evidence	Caveat
$ \delta_M < 2\%$ at 2:1	4 domains, bootstrap CIs, all pass	Requires near-stationarity
Multi-tap: 94.8% pass	213 measurements, 8 SBP trees	Failures = Solar (fast) or Nuclear (break)
Subcomp EITT	Fossil, Financial subsets pass	Renewables-only fails (-4.5%)
EITT Inversion	K=2 fails, K=4 passes (gold/silver)	Bidirectional: upward + downward confirmed
Adaptive decimation	Gold/Silver 10:1: +1.39% fixed -> -0.43% adaptive	Adds hyperparameters
Renyi generalisation	Confirmed $q=0.1$ to 5.0, all near-invariant	Shannon is not special. Lives in geometry.

LAYER 3 — OPEN PROBLEMS (for the CoDa community)

Problem	Description
O-1: Formal bound	Prove $ \delta_M \leq f(D, \sigma_A^2, \rho_1, M)$ for stationary processes
O-2: Variance decomp	Is the variance reduction factor determined by temporal dependence alone?
O-3: Fisher-Rao	Curvature interpretation of EITT residual under Fisher-Rao metric?
O-4: Subcomp conditions	When does full-comp EITT imply subcompositional EITT?
O-5: Adaptive optimality	Optimal $M^*(t)$ minimising $ \delta $ subject to compression constraint?

5. Empirical Scorecard

Domain	D	N	delta 2:1	95% CI	Status
Gold/Silver prices	2	339	+0.085%	[-0.19%, +0.16%]	PASS
Energy World	7	25	-0.589%	[-0.97%, +0.20%]	PASS
Financial (price-level)	9	74	+0.004%	[+0.004%, +0.006%]*	PASS*
Energy Germany	7	36	+0.029%	[-0.12%, +0.09%]	PASS

*Financial CI excludes zero (tiny systematic +0.004% bias). Not market-cap weighted.

6. Honest Disclosures

#	Disclosure
1	EITT is an empirical observation, NOT a theorem. Formal proof is Open Problem O-1.
2	Geometric-mean superiority over arithmetic mean NOT empirically demonstrated at low compression.
3	Spatial EITT has been RETRACTED (HEALPix NESTED ordering error).
4	Financial = mean closing prices (price-level, K=9), NOT market-cap.
5	$d(\text{CoDa})/dt$ is operationalised but not formally optimised.
6	EITT Inversion: bidirectional confirmed but needs more domains.
7	Adaptive decimation adds hyperparameters; cannot rescue fully non-stationary data.
8	Aitchison distances EXPAND under decimation. EITT is NOT contraction/smoothing.

7. Our Position in One Paragraph

Shannon entropy of a compositional time series appears empirically near-invariant under geometric-mean temporal decimation, provided the process is near-stationary. This holds across the full Renyi family ($q = 0.1$ to 5.0), confirming the phenomenon lives in the Aitchison geometry, not in a special property of Shannon entropy. ILR balance means are exactly preserved (mathematical, not empirical) regardless of SBP tree choice. When the invariance fails, the failure is diagnostic: upward inversion identifies hidden structure, downward inversion identifies non-stationarity. The balance velocity $d(\text{CoDa})/dt$ serves as the control signal that determines the safe compression window. The formal proof bounding the residual in terms of Aitchison variance and temporal dependence is an open problem we bring to the CoDa community.

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All code, data, and test results: github.com/PeterHiggins19/Higgins-Unity-Framework

Multi-AI adversarial review (Claude, ChatGPT, Grok, Gemini, Copilot) applied throughout.